

Benchmark Methodology

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Most existing quantum transpilers and AI routing systems are not fully hardware-aware, which can lead to higher SWAP counts, larger circuit depth, and lower execution reliability on real quantum hardware.

This benchmark was designed to test how well different systems optimize quantum circuits while considering real hardware connectivity and noise constraints. All methods were evaluated on the same IBM heavy-hex hardware, using identical circuits, qubit counts, and random seeds to ensure a fair comparison.

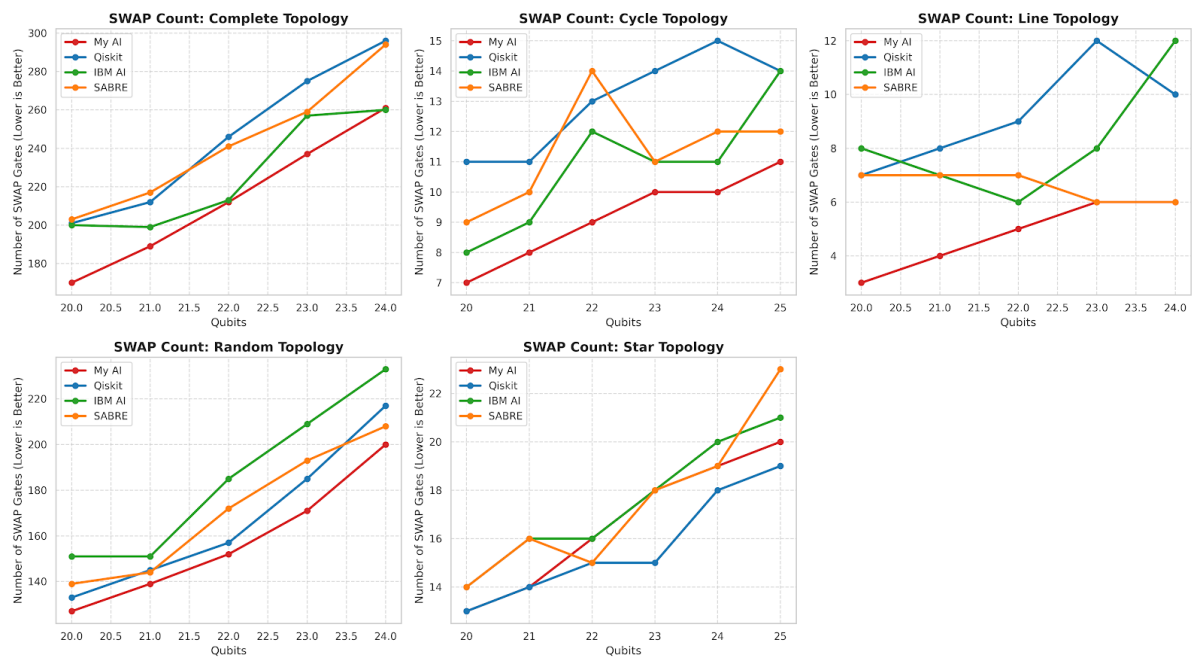
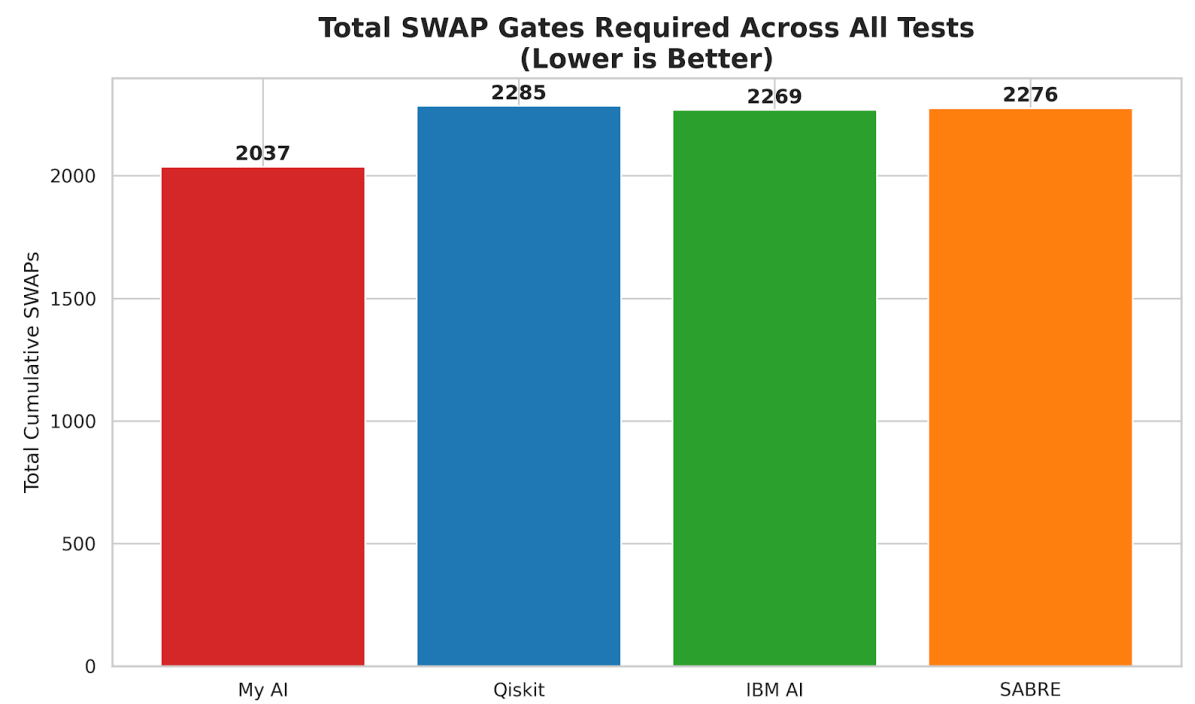
The results compare SWAP count, circuit depth, and Estimated Success Probability (ESP) to measure routing efficiency and hardware execution quality.

- All transpilers were tested on the exact same logical circuits, graph topologies, qubit counts, hardware topology, and random seeds.
- Benchmarks were run on a live IBM heavy-hex hardware slice extracted from `ibm_fez` using real calibration data.
- A 30-qubit high-fidelity hardware subgraph, originally designed as part of the custom compiler's hardware-aware optimization pipeline, was reused uniformly across all transpilers so that every system operated under the exact same live connectivity and noise constraints.
- The benchmark covered five topology families:
 - line
 - cycle
 - star
 - random
 - complete
- Logical qubit sizes ranged from 20 to 25 qubits.
- Random and complete graphs were included because they create the hardest routing cases on sparse heavy-hex hardware.
- Every test used the same evaluation seeds:
 - 42
 - 123
 - 456
- Compilers were run under their standard high-optimization settings:
 - Qiskit at optimization level 3
 - IBM AI routing in its AI optimization mode
 - SABRE with multiple layout and swap trials and decay-based routing

- All circuits were fully routed and legalized onto the same hardware coupling constraints before scoring.
- Performance was measured using three metrics:
 - SWAP count
 - circuit depth
 - Estimated Success Probability (ESP)
- ESP was computed from the live hardware noise model using:
 - readout errors on the physical qubits actually used
 - two-qubit gate errors on all routed entangling operations
 - extra penalty for SWAP decomposition, since each SWAP expands into multiple two-qubit interactions
- The same ESP formula was applied to every compiler, with no special treatment or postprocessing.
- Benchmark problems were reconstructed directly from stored experiment configurations so that every compiler was evaluated on identical routing instances.
- No topology-specific tuning, manual corrections, or compiler-specific postprocessing were applied outside the standard transpilation pipeline.

In a small number of cases, slightly higher SWAP count or circuit depth was intentionally accepted as a tradeoff to achieve higher Estimated Success Probability (ESP), since routing through lower-noise qubits and higher-fidelity couplings can produce more reliable execution on real hardware even if the circuit is not strictly the shortest possible.

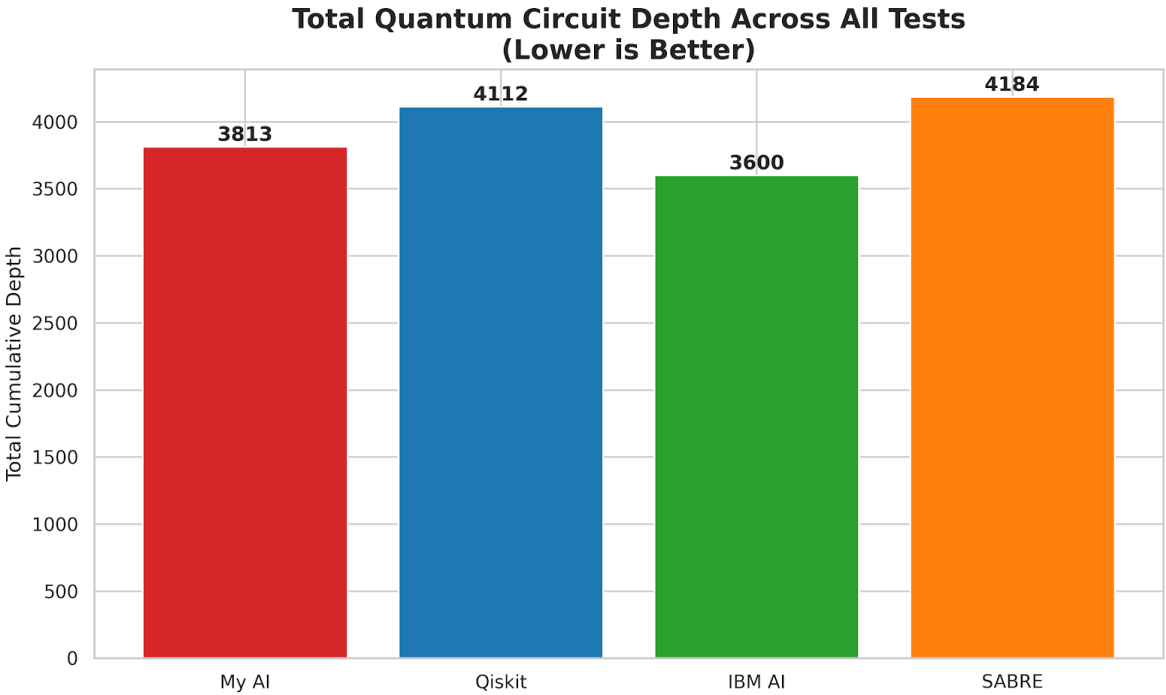
SWAPS COUNT

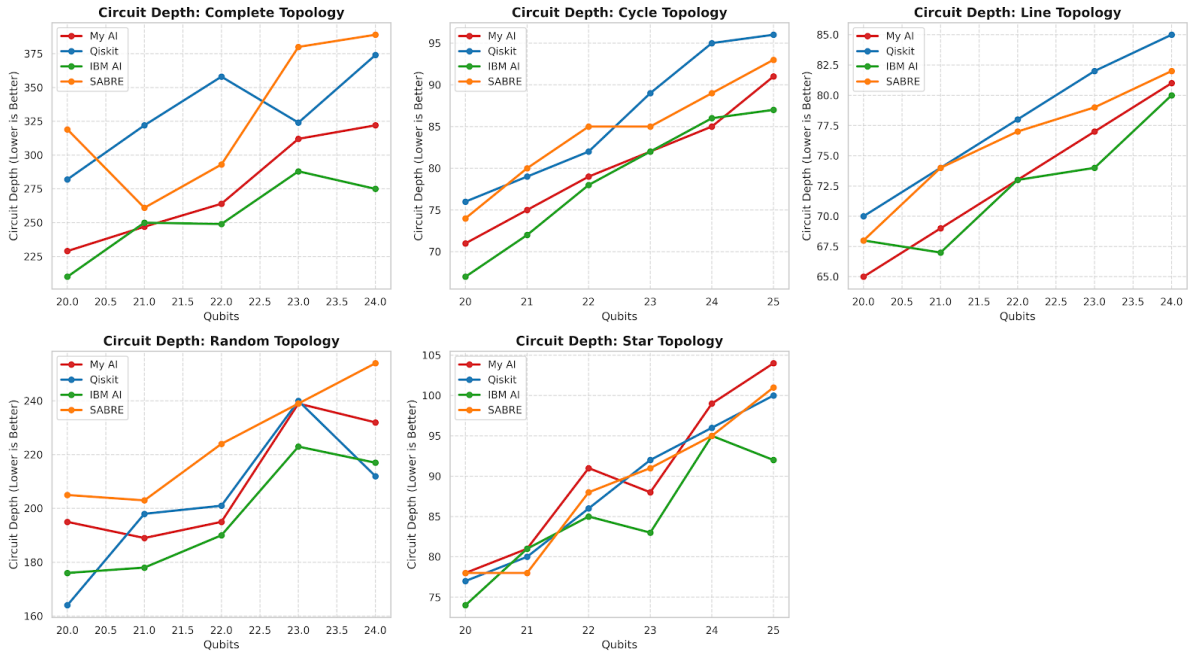


Topology	Qubits	My AI SWAPS	Qiskit SWAPS	IBM AI SWAPS	SABRE SWAPS
Complete	20	170	201	200	203
Complete	21	189	212	199	217
Complete	22	212	246	213	241
Complete	23	237	275	257	259
Complete	24	261	296	260	294
Cycle	20	7	11	8	9

Cycle	21	8	11	9	10
Cycle	22	9	13	12	14
Cycle	23	10	14	11	11
Cycle	24	10	15	11	12
Cycle	25	11	14	14	12
Line	20	3	7	8	7
Line	21	4	8	7	7
Line	22	5	9	6	7
Line	23	6	12	8	6
Line	24	6	10	12	6
Random	20	127	133	151	139
Random	21	139	145	151	144
Random	22	152	157	185	172
Random	23	171	185	209	193
Random	24	200	217	233	208
Star	20	13	13	14	14
Star	21	14	14	16	16
Star	22	16	15	16	15
Star	23	18	15	18	18
Star	24	19	18	20	19
Star	25	20	19	21	23

DEPTH COUNT



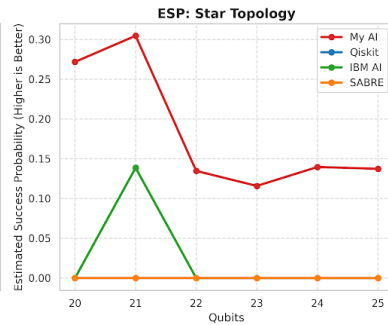
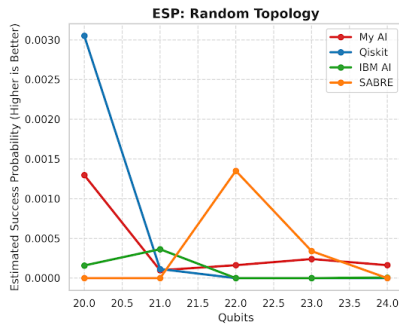
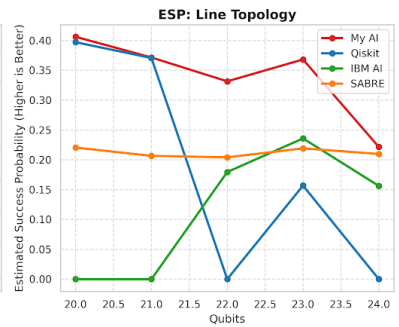
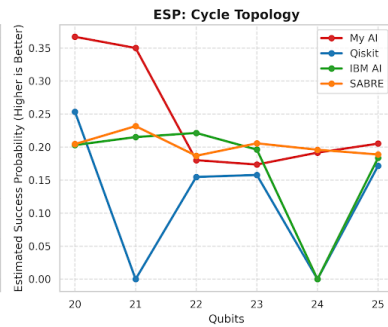
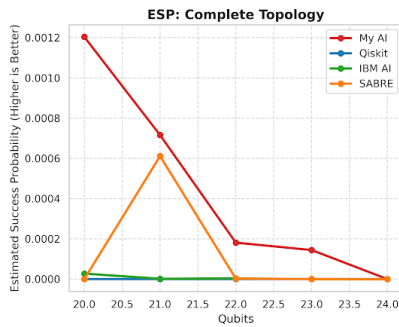
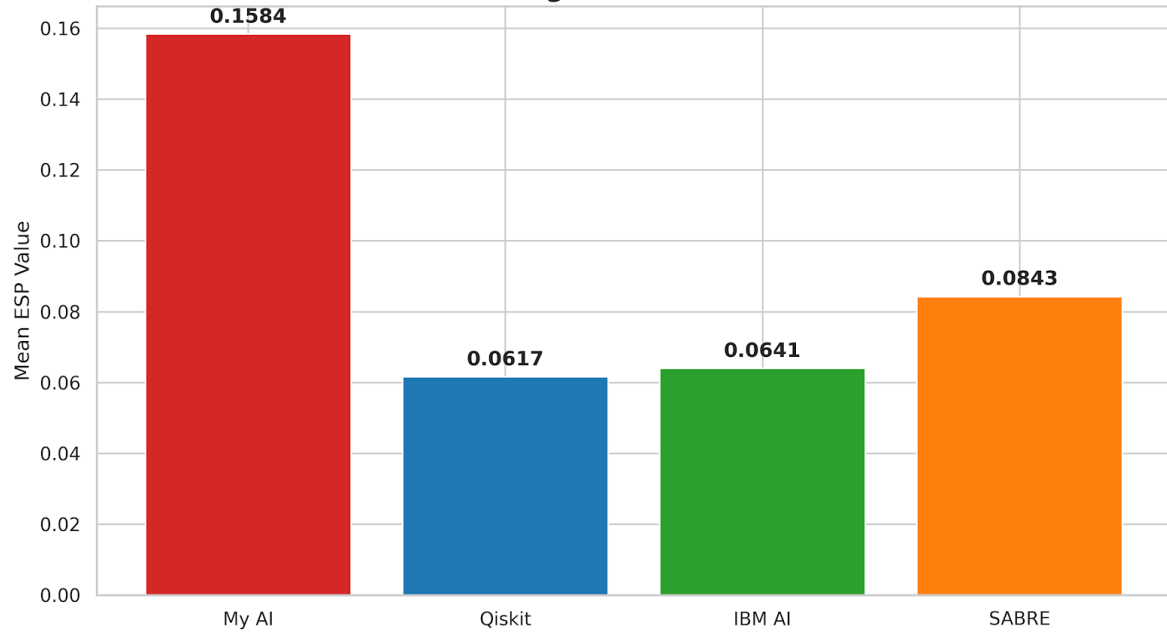


Topology	Qubits	My AI DEPTH	Qiskit DEPTH	IBM AI DEPTH	SABRE DEPTH
Complete	20	229	282	210	319
Complete	21	247	322	250	261
Complete	22	264	358	249	293
Complete	23	312	324	288	380
Complete	24	322	374	275	389
Cycle	20	71	76	67	74
Cycle	21	75	79	72	80
Cycle	22	79	82	78	85
Cycle	23	82	89	82	85
Cycle	24	85	95	86	89
Cycle	25	91	96	87	93
Line	20	65	70	68	68
Line	21	69	74	67	74
Line	22	73	78	73	77
Line	23	77	82	74	79
Line	24	81	85	80	82
Random	20	195	164	176	205
Random	21	189	198	178	203
Random	22	195	201	190	224
Random	23	239	240	223	239
Random	24	232	212	217	254
Star	20	78	77	74	78
Star	21	81	80	81	78
Star	22	91	86	85	88
Star	23	88	92	83	91
Star	24	99	96	95	95

Star	25	104	100	92	101
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ESP COUNT

**Average Estimated Success Probability (ESP) Across All Tests
(Higher is Better)**



Topology	Qubits	My AI ESP	Qiskit ESP	IBM AI ESP	SABRE ESP
Complete	20	0.00120449	0	0.00002771	0
Complete	21	0.00071624	0	0.00000214	0.00061161
Complete	22	0.00018151	0	0.00000412	0.00000264
Complete	23	0.00014489	0	0.00000024	0.00000087

Complete	24	0.00000030	0	0	0
Cycle	20	0.367011	0.253598	0.203073	0.204600
Cycle	21	0.350089	0	0.215169	0.231761
Cycle	22	0.180226	0.154802	0.221433	0.186893
Cycle	23	0.173584	0.157826	0.196216	0.205739
Cycle	24	0.191641	0	0	0.196023
Cycle	25	0.205277	0.171706	0.183780	0.188820
Line	20	0.406226	0.397207	0	0.220540
Line	21	0.371619	0.370661	0	0.206726
Line	22	0.331605	0	0.179580	0.204199
Line	23	0.368257	0.156874	0.235936	0.219257
Line	24	0.221634	0	0.156297	0.209883
Random	20	0.00129793	0.00305152	0.00015928	0
Random	21	0.00009943	0.00011498	0.00036432	0
Random	22	0.00016311	0	0	0.00135114
Random	23	0.00024065	0	0	0.00034032
Random	24	0.00016309	0	0.00000742	0
Star	20	0.271813	0	0	0
Star	21	0.304654	0	0.138738	0
Star	22	0.134798	0	0	0
Star	23	0.115936	0	0	0
Star	24	0.139745	0	0	0
Star	25	0.137408	0	0	0